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Arctic Heatwaves Could Significantly Influence the Isoprene Emissions From Shrubs

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Key Points:

- The hourly temperature response curve of Arctic *Salix* spp. is similar to temperate plants and is well represented by the Model of Emissions of Gases and Aerosols from Nature (MEGAN) model
- The isoprene capacity of willows exhibited a pronounced response to the mean temperature of the previous day
- MEGAN predicts a 66% higher isoprene emission for Arctic willows during an Arctic heatwave after incorporating the findings of this study

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Warming climate in the Arctic is leading to an increase in isoprene emission from ecosystems. We assessed the influence of temperature on isoprene emission from Arctic willows with laboratory and field measurements. Our findings indicate that the hourly temperature response curve of *Salix* spp., the dominant isoprene emitting shrub in the Arctic, aligns with that of temperate plants. In contrast, the isoprene capacity of willows exhibited a more substantial than expected response to the mean ambient temperature of the previous day, which is much stronger than the daily temperature response predicted by the current version of the Model of Emissions of Gases and Aerosols from Nature (MEGAN). With a modified algorithm from this study, MEGAN predicts 66% higher isoprene emissions for Arctic willows during an Arctic heatwave. However, despite these findings, we are still unable to fully explain the high temperature sensitivity of isoprene emissions from high latitude ecosystems.

Plain Language Summary Isoprene plays an important role in atmospheric chemistry and climate. As the climate warms, ecosystems release more isoprene into the air due to higher temperatures and changes in plant species composition. The Arctic is a hotspot for climate change. In this study, we investigated how temperature affects isoprene emissions from Arctic willow shrubs by conducting experiments in both the lab and the field. Our results show that the response of *Salix* spp. (willows) in the Arctic react to short-term temperature changes is similar to plants in temperate regions. Interestingly, the amount of isoprene that willows release responds strongly to the average temperature of the previous day. This response is much stronger than predicted by models. Considering the insights from our study, isoprene emissions from Arctic willows are much higher during Arctic heatwaves than previously thought. This study can help us understand how isoprene from Arctic willows will affect atmospheric composition and climate and contribute to corresponding feedbacks on the local climate.

1. Introduction

Rapid climate change in the Arctic is strongly influencing the local ecosystems (Box et al., 2019; Kim et al., 2020; Rantanen et al., 2022), and the change in both climate and ecosystems can alter the emissions of biogenic volatile organic compounds (BVOCs) and atmospheric chemistry in the Arctic (Faubert et al., 2010; Lindwall, Schollert, et al., 2016; Rinnan et al., 2020). Isoprene is the most abundant highly reactive BVOC emitted globally and also in the Arctic (Guenther et al., 2012; Rinnan et al., 2020). The changes in isoprene emissions could affect the formation and characteristics of secondary organic aerosol (SOA), ozone, the lifetime of methane, and thus the climate system in the Arctic (Boy et al., 2022; Kulmala et al., 2004; Petäjä et al., 2022; Tunved et al., 2006; Weber et al., 2022). The emission of isoprene is controlled by environmental conditions including temperature and solar radiation (Guenther et al., 1993; Monson et al., 1992; Tingey et al., 1979), thus a rapidly warming climate in the Arctic is favorable for increasing emission of isoprene (Bauwens et al., 2018; Emmerson et al., 2020; Kramshøj et al., 2016; Lindwall, Svendsen, et al., 2016; Seco et al., 2020). Additionally, climate change is extending the growing season and shifting plant productivity and community composition (Park et al., 2016; Valolahti et al., 2015; Wang et al., 2020), which will also affect BVOC emissions indirectly (Rinnan et al., 2020). Specifically, the expansion of shrubs into graminoid tundra systems ("shrubification") (Mekonnen et al., 2021) is a potential driver of isoprene emission change.

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The shrub willow (*Salix* spp.) is a prevalent source of isoprene in high latitude ecosystems (Li et al., 2023; Potosnak et al., 2013; Simin et al., 2021). Recent whole-ecosystem measurements suggest that the temperature response of isoprene emissions in high-latitude tundra ecosystems is stronger than what current isoprene emission models predict (Angot et al., 2020; Holst et al., 2010; Li et al., 2023; Seco et al., 2020, 2022; Selimovic et al., 2022; Vettikkat et al., 2023). In contrast, leaf/branch-level studies showed that the hourly temperature response of Arctic willow species (*Salix pulchra* Cham. and *S. myrsinites* L.) is similar to that of temperate plants (Li et al., 2023; Potosnak et al., 2013). Besides the current temperature, it is known that basal isoprene emission capacity can also acclimate to the temperature and light of the past day or more (Geron et al., 2000; Hanson & Sharkey, 2001; Sharkey et al., 1999; Wiberley et al., 2008). Ground chamber experiments have qualitatively demonstrated that long-term warming can increase isoprene emissions from high latitude ecosystems (Lindwall, Svendsen, et al., 2016; Tiiva et al., 2008; Valolahti et al., 2015). However, few data are available for quantifying the acclimation to warming and heatwaves in high latitudes. In this study, we designed laboratory and field experiments to investigate the temperature response, on time scales of minutes to days, of isoprene emissions from Arctic *Salix* spp. This finding was integrated into the Model of Emissions of Gases and Aerosols from Nature (MEGAN) and used to investigate isoprene emissions from high latitude shrubs under warming conditions.

2. Materials and Methods

The measurements were conducted as part of the Biogenic Emissions and Aerosol Response on the North Slope (BEAR-oNS) study during the summers of 2022–2023 on the North Slope, Alaska. The vegetation (*Salix* spp.) and volatile organic compound (VOC) samples were collected around the Toolik Field Station (TFS, 68° 38′N, 149° 36′W). Isoprene emissions from three local *Salix* spp. (*Salix pulchra*, *S. glauca* and *S. reticulata*, see Figure S1 in Supporting Information S1) were measured. Two types of experiments were designed to assess the impact of short-term and long-term temperature changes on isoprene emissions from *Salix* spp. For the short-term experiments, plant samples were detached and brought to the lab at TFS. Long-term temperature experiments were conducted in the field on two separate *S. glauca* bushes. VOC samples were collected using sorbent cartridges (Tenax TA and Carbograph 5TD; Markes International, UK). The samples were capped, refrigerated, and then shipped back to the lab at the University of California, Irvine for analysis. The time between sample collection and analysis was about 2 weeks.

2.1. Temperature Response Curve Experiments

The short-term temperature response experiment aimed to assess response of isoprene emission to temperature changes on a time scale of minutes to hours. A portable custom-made leaf glass chamber system was used for these experiments. The system details are described by Nagalingam et al. (2023). Branches were cut, placed in water, and exposed to a natural light diurnal cycle 1–2 days before each experiment. The duration of darkness ranged from 1 to 5 hr at TFS during the campaign. One chamber blank VOC sample (from an empty chamber) was collected at the beginning of every experiment before putting the plant into the glass chamber. After placing a leaf into the glass chamber, VOC sampling was initiated only after the photosynthesis rate stabilized. The photosynthetic photon flux density (PPFD) is approximately 1,000 μmol m^{-2} s^{-1} in the chamber. In 2022, for *S. glauca* ($n = 3$) and *S. pulchra* ($n = 4$), leaf temperature was increased from 10°C to 45°C in 5°C steps. In 2023, for *S. reticulata* ($n = 4$), the temperature was increased from 20°C to 40°C in 5°C steps. Each step lasted 1 hour (except 20°C, which was 2 hr). Samples were taken using sorbent cartridges (see next section) during the last 10–15 min of the hour with a flow rate of 200 cc min^{-1} for 5 min, yielding a 1 L VOC sample.

2.2. Acclimation Experiments

The acclimation experiments focused on investigating the influence of the temperature of the past several days on isoprene emission capacity. Isoprene emissions from seven mature *S. glauca* leaves were measured over 21 days, from 16 July to 5 August 2023. The mean July temperature at TFS was 10.4°C from 2020 to 2023. During our measurement period, there were 5 days (July 21–24, and August 5) when the daily temperatures exceeded the 95th percentile of the daily temperature records of 16.5°C for the same period (2020–2023). VOC measurements were taken using an LI-6400XT portable photosynthesis system (LI-COR Biosciences, USA) with an inlet flow rate of 730 cc min^{-1} (Figure S2 in Supporting Information S1). All measurements occurred between 13:00 and

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17:00 local time to mitigate potential diurnal effects. A chamber blank VOC sample was collected every measurement day from an empty LI-6400XT chamber before measurements. Leaves were placed in the chamber at a fixed 20°C temperature with a PPFD intensity of 1000 μmol m^{-2} s^{-1}. A 1L VOC sample was collected at a flow rate of 200 cc min^{-1} for 5 min after the leaves had been in the chamber for 30 min. The meteorology data used in this study were measured at TFS and are available through the MesoWest Database (https://mesowest.utah.edu/).

2.3. GC-TOF-MS

The sampled sorbent cartridges were transported to our laboratory at the University of California, Irvine, where they were thermally desorbed using a TD autosampler (Ultra-xr; Markes International). The desorbed VOCs were injected into a gas chromatograph (GC) (7890B; Agilent Technologies, CA, USA) equipped with a 60 m Rxi-624Sil MS capillary column (Restek, PA, USA). The column eluate was channeled to an electron impact ionization time-of-flight mass spectrometer (BenchTOF-Select; Markes International) and a flame ionization detector (FID, Agilent) for compound identification and quantification. Detailed explanation of the GC methodology including the GC oven temperature program, calibration protocols, and measurement uncertainties are described by Nagalingam et al., 2022.

2.4. MEGAN Model

The isoprene flux in MEGAN version 3 is calculated as:

$$F = \varepsilon \cdot \mathrm{LAI} \cdot \gamma_T \cdot \gamma_P \cdot \gamma_A \cdot \gamma_C \cdot \gamma_{\mathrm{SM}}, \tag{1}$$

where ε and LAI represent the leaf-level standard emission factor (μmol m^{-2} s^{-1})) and leaf area index (LAI, m^2 m^{-2}), respectively. γ_T, γ_P, γ_A, γ_C, and γ_{SM} represent the emission activity factors of isoprene emissions for temperature, light condition, leaf age, CO_2 and water stress, respectively. The emission factor (ε) is defined as the rate of isoprene emission when the leaf temperature is 30°C under a PPFD of 1,000 μmol m^{-2} s^{-1}. We will focus on the γ_T in this study, and the details about other factors can be found in previous publications (Guenther et al., 2006, 2012; Wang et al., 2022).

MEGAN considers the effects of the current temperature and the long-term (1–10 days) temperature on isoprene emission. The default temperature response algorithm for isoprene in MEGAN is:

$$\gamma_T = E_{\mathrm{opt}} \cdot \frac{C_{T2} \cdot e^{C_{T1} \cdot x}}{C_{T2} - C_{T1} \cdot (1 - e^{C_{T2} \cdot x})}, \tag{2}$$

where x is:

$$x = \frac{1}{0.00831} \cdot \left(\frac{1}{T_{\mathrm{opt}}} - \frac{1}{T} \right). \tag{3}$$

T is the leaf temperature. C_{T1} and C_{T2} are both empirical coefficients. E_{opt} and T_{opt} are:

$$T_{\mathrm{opt}} = 40 + 0.6 \cdot (T_{240} - 24) \tag{4}$$

$$E_{\mathrm{opt}} = 2 \cdot e^{0.05 \cdot (T_{24} - 24)} \cdot e^{0.05 \cdot (T_{240} - 24)}, \tag{5}$$

where T_{24} (°C) and T_{240} (°C) denote the mean air temperature in previous 24 and 240 hr, respectively. T_{opt} (°C) and E_{opt} represent the impact of long-term temperatures on the optimal temperature and the shape of the temperature response curve.

3. Results and Discussion

3.1. The Short-Term Temperature Response of Isoprene

Our chamber experiments show that the short-term temperature response of isoprene from Arctic willows (*Salix* spp.) is consistent with the MEGAN model (Figure 1), with isoprene emission increasing with temperature, reaching an optimal level (point of peak emission) around 40°C before declining, following typical enzyme

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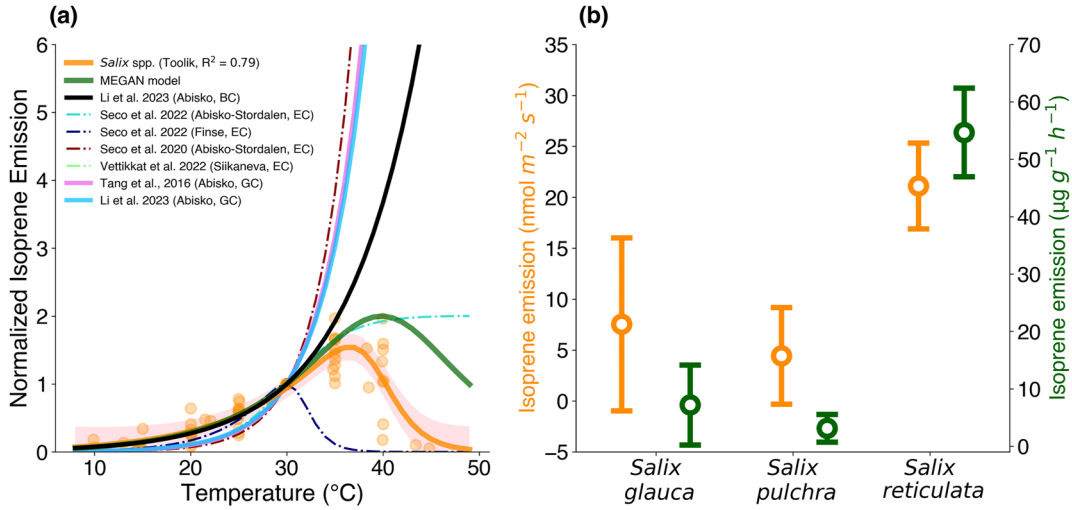

Figure 1. (a) Comparing temperature responses of isoprene emissions between this and previous studies. Short-term temperature response curve of willows (orange solid line) derived through leaf chamber experiments in this study, along with tundra whole-ecosystem measurement response curves from previous studies (various colors and patterns). The orange shadow represents the 95% confidence intervals. GC, BC, and EC denote ground chamber, branch chamber, and eddy-covariance measurements. Curves are normalized to emission at a leaf temperature of 30°C. (b) Emission factors of different *Salix* spp. in the Arctic. Emission factor is defined as isoprene emission capacity at 30°C and PPFD of 1,000 μmol m^{-2} s^{-1}. Averaged emission factors are shown per leaf area (left axis, orange) and per dry leaf mass (right axis, green). Points and error bars represent mean and standard deviation of emission factors.

behavior of increasing activity followed by denaturation (Guenther et al., 1993). The parameters for the temperature response curve of *Salix* spp. were fitted using the Arrhenius equation format presented in Equations 2 and 3, with a comparison of the empirical parameters for the MEGAN default and *Salix* spp. models provided in Table S1 in Supporting Information S1. The fitted curves for individual samples are presented in Figure S3 in Supporting Information S1. The optimal temperature for isoprene emission is known to be influenced by the temperature of the past 1–10 days, based on measurements of temperate plants (Guenther et al., 2012; Papiez et al., 2009). Arctic willows that grow in a cold environment are expected to have a lower optimal temperature (36°C) than those growing in a temperate climate. Therefore, the model fitted for willows (*Salix* spp.) has a different optimal temperature (Figure 1) than that represented in the current MEGAN model which was based primarily on observations of woody temperate plants (Guenther et al., 2012).

We also compare the short-term temperature response curve of isoprene emissions with observations in the Arctic ecosystem reported by previous studies. We found our temperature response curve is close to the curve derived from the branch chamber experiment by Li et al. (2023) for *S. myrsinites*, and the difference at high temperatures (>35°C) is caused by the equation formats that were used to fit the line. In addition, Potosnak et al. (2013) found that their temperature curve experiments for *S. pulchra* reproduced the temperature curve of the MEGAN model. Therefore, we conclude that the temperature curve of willows in the Arctic is consistent with the temperature response in the current MEGAN model, though with a lower optimal temperature. However, the temperature curve of willows is not consistent with the whole ecosystem measurements of isoprene from the high-latitude regions (e.g., Seco et al., 2022; Tang et al., 2016; Vettikkat et al., 2022). The temperature curve derived from the whole ecosystem measurements showed the exponential increase regime and did not exhibit a turning point because it is rare for leaf temperature to reach the optimal level in a high-latitude environment. MEGAN can capture the variability in isoprene emissions in temperate and tropical ecosystems as measured using eddy-covariance techniques (Potosnak et al., 2014; Sarkar et al., 2020). However, it does not account for measurements from high latitudes (Seco et al., 2022; Vettikkat et al., 2022), indicating that discrepancies are not due to measurement methodologies. We also investigated the impact of acclimation of isoprene emission factors to ambient temperature in Section 3.3, but this alone cannot fully explain the differences.

We also observed significant variation in emission factors among different *Salix* spp. ((b) in Figure 1). One dwarf willow species, *S. reticulata*, had a notably higher emission factor (21.1 ± 4.2 nmol m^{-2} s^{-1}) compared to *S. glauca* (7.5 ± 8.5 nmol m^{-2} s^{-1}) and *S. pulchra* (4.4 ± 4.7 nmol m^{-2} s^{-1}). The differences are even greater

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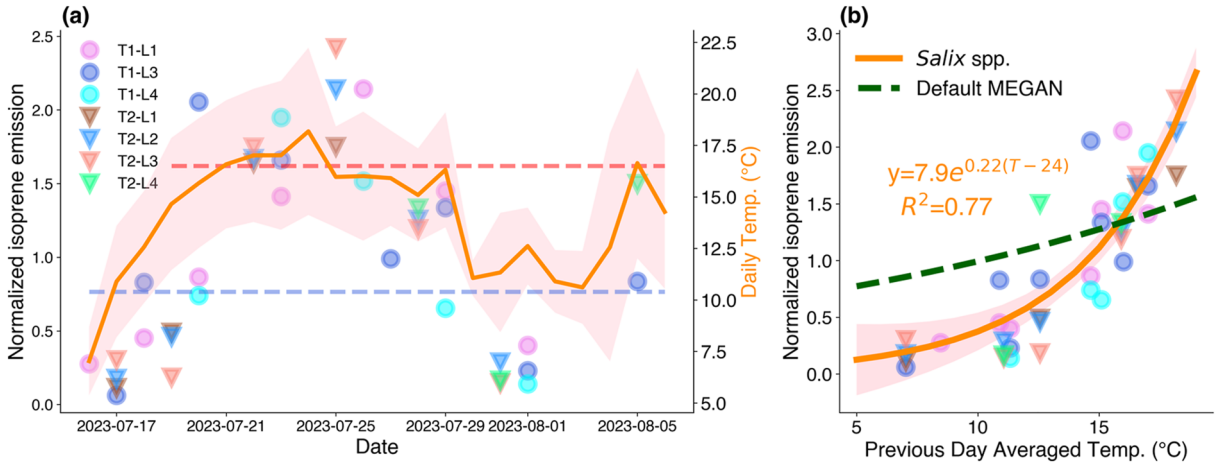

Figure 2. (a) The time-series of normalized emission factors from different leaves (various colors and patterns, left axis) and daily temperatures (solid orange line, right axis). The blue and red dashed lines represent the mean daily temperature and 95th percentile of the daily temperature records during 2020–2023, respectively. The orange shadow represents the standard deviation of daily temperature in (a). The emission factors were normalized by dividing the mean emission factors for individual leaves. (b) The correlation between normalized emission factors and the previous-day averaged temperature is shown alongside that in the default MEGAN model (dashed green line). The orange shadow represents the 95% confidence intervals in (b). The equations for the fitted lines in (b) are also presented.

when the emission factor is normalized by leaf mass indicating that the species with lower emissions have denser leaves. Potosnak et al. (2013) reported an averaged emission factor of 12.4 (±10.6) nmol m^{-2} s^{-1} for *S. pulchra*, and Simin et al. (2021) reported an averaged emission factor of 10.4 (±6.1) nmol m^{-2} s^{-1} for *S. myrsinites*. In addition to the significant variation among species, environmental temperature could also affect the emission factors. The difference of emission factors for *S. pulchra* between our measurements and those from Potosnak et al. (2013) could be explained by the acclimation of plants to the previous day temperature, which is presented in the next section.

3.2. The Acclimation of Isoprene Emission Factors to the Temperature

Our field measurements show that isoprene emission factors, made under almost identical light and leaf temperature conditions, change with the daily ambient temperature variations as seen in (a) in Figure 2. The highest isoprene emissions occurred during the warmest period of our measurements. Furthermore, the long-term temperature response experiments show that the emission capacity of *S. glauca* increases exponentially with the mean air temperature averaged over the past one day ((b) in Figure 2). The emissions were measured at a leaf temperature of ~20°C and then converted to the corresponding value at 30°C, the standard temperature in MEGAN, using the temperature curve presented for *Salix* spp. in Figure 1. The emission factors were normalized by dividing the mean emission factors for individual leaves. The analysis using the original data at 20°C yields the same conclusion (Figure S4 in Supporting Information S1). We also tested other factors including the mean of temperature, PPFD, VPD (Vapor Pressure Deficit), and the product of PPFD and temperature for the previous 1–240 hr before the measurement (Figure S5 in Supporting Information S1). The mean air temperature over the preceding 35 hr has the highest Pearson's correlation coefficient of 0.88 ($p < 0.001$) with the log of emission factors, which is close to Pearson's correlation coefficient for the mean temperature of the previous day. It is known that past growth temperature affects isoprene emission capacity of plants (Wiberley et al., 2008) through the accumulation of enzymes and substrates (Grote & Niinemets, 2008). Isoprene emission can acclimate to a new temperature and light condition within 5–30 hr (Hanson & Sharkey, 2001), and a previous study found that the basal level isoprene emission of oak trees was highly correlated with the average temperature of the previous 2 days (Hanson & Sharkey, 2001). Our data shows that for S. glauca the emission factors are more related to the mean temperature of the preceding day.

3.3. MEGAN Simulations

The acclimation mechanism has been included in the current MEGAN model by using an exponential relationship as presented in Equation 5 (Guenther et al., 2012), and the validation of flux measurements shows that the

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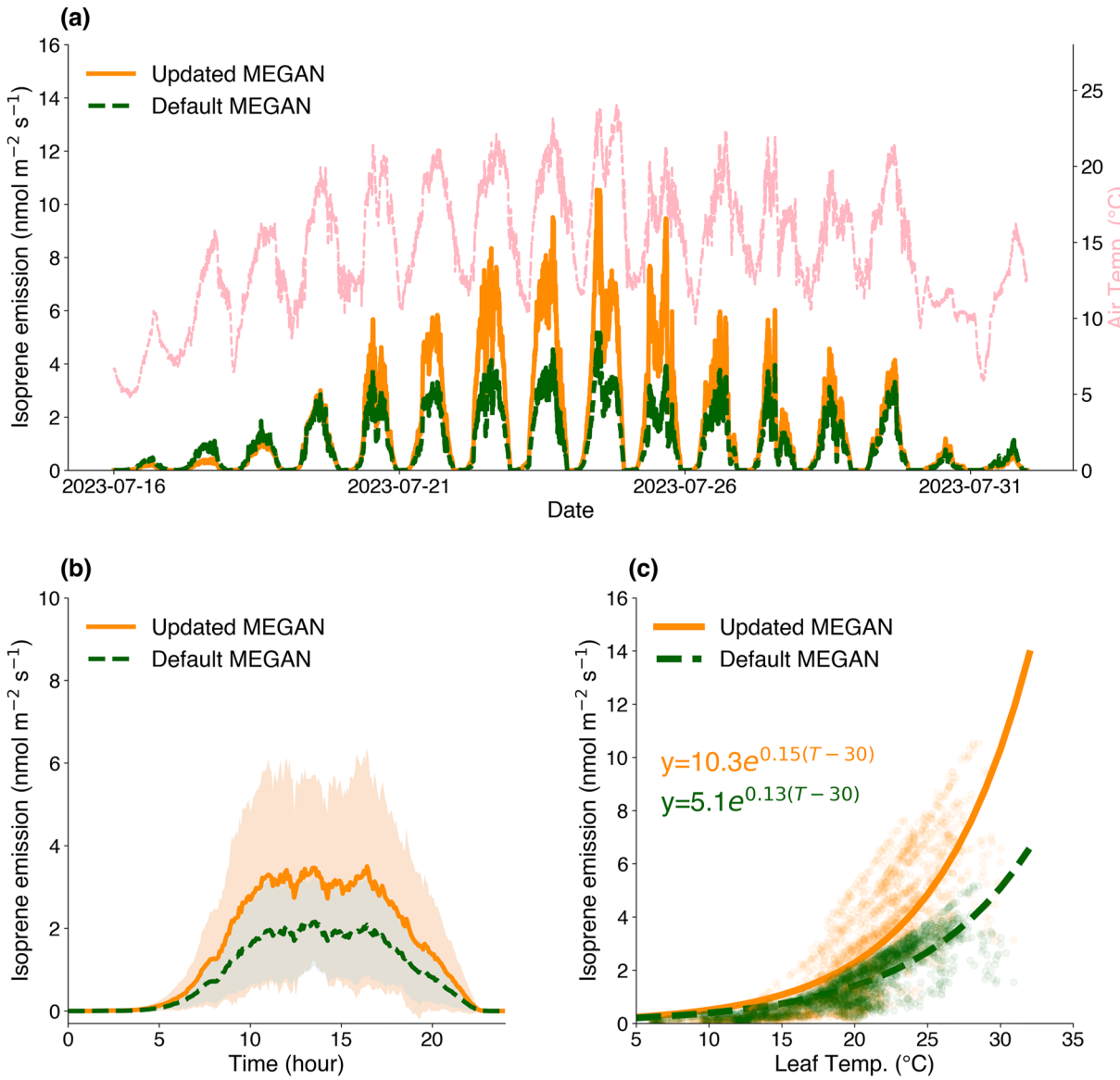

Figure 3. The time-series of simulated isoprene emissions from the default (dashed green line, left axis) and updated MEGAN models (solid orange line, left axis), alongside air temperature (dashed pink line, right axis), during the heatwave period (July 16–August 1) in 2023 at the Toolik Field Station (a). The diurnal cycle of simulated isoprene emissions shown in (a) is depicted in (b), and the shadows in (b) represents standard deviations of isoprene emission. (c) Presented the short-term temperature response curve of isoprene flux, obtained from flux estimations using both the default and updated MEGAN models. The equations for the fitted lines in (c) are also presented.

acclimation mechanism can improve model performance (e.g., Potosnak et al., 2014). However, our results indicate that the emission factors of *S. glauca* have a heightened temperature sensitivity to the mean temperature of the preceding day compared to the current, referred to here as the default, model ((b) in Figure 2). We updated Equation 5 in MEGAN with the equation derived from this study as:

$$E_{\text{opt}} = 7.9 \cdot e^{0.22\cdot(T_{24}-24)}, \tag{6}$$

where T_{24} (°C) denotes the mean air temperature of the preceding day. We estimated the isoprene flux from a unit area (1 m^2) of *S. glauca* with LAI of 1.5 m^2 m^{-2} at TFS during the heatwave period (July 16–August 1) in 2023. The higher temperature sensitivity of the updated MEGAN model results in lower emissions during cooler days but higher emissions during warmer days, as shown in (a) in Figure 3. Additionally, the diurnal cycle reveals that the updated MEGAN model has higher emissions than the default MEGAN model with higher emission

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factors ((b) in Figure 3) during warm periods, and we found that isoprene flux could increase by 66% by updating the acclimation mechanism of isoprene (Figure 3). Consequently, the existing model is likely to considerably underestimate the isoprene emissions from willows during warm periods and overestimate during cold periods. In addition, the isoprene emissions from Arctic willows could undergo a substantial increase in response to the rapid warming in the Arctic region.

To compare with whole ecosystem flux measurements, we constructed the temperature response curve of *S. glauca* based on results from both default and updated MEGAN models during the heatwave ((c) in Figure 3). The temperature curves were fitted using the format of the exponential equation:

$$F = F_0 \cdot e^{\beta \cdot (T - T_{\text{std}})}, \tag{7}$$

where T_{std} (=30°C) is the standard leaf temperature in MEGAN. F_0 and β are the empirical parameters.

The results show that acclimation of emission capacity shifted the averaged emission factor (F_0) from 5.1 to 10.3 nmol m^{-2} s^{-1} and increased the temperature sensitivity of isoprene flux, with β shifting from 0.13 to 0.15. However, this value is still considerably lower than the values of 0.17–0.23 observed for other studies (Li et al., 2023; Seco et al., 2020, 2022; Tang et al., 2016; Vettikkat et al., 2022). Isoprene emission at some of these sites are not dominated by the contribution (percent cover) of *Salix* spp. (Seco et al., 2022; Vettikkat et al., 2022) and so other isoprene emitters, like sedges (Ekberg et al., 2009), are likely to be responsible for the high temperature sensitivity of isoprene observed in ecosystem studies in the Arctic.

4. Conclusions

The isoprene emission response of Arctic shrub willows to temperature change was assessed in this study through field and laboratory measurements. We found the temperature response, on time scales of minutes to days, of Arctic willows is consistent with that represented in the current MEGAN model, which was based primarily on observations with woody temperate plants. In addition, we found that the isoprene emission capacity is acclimated to the averaged temperature of the previous day and that response is stronger than that predicted in MEGAN. However, findings in this study still cannot explain the highly temperature sensitive response curves derived from whole-ecosystem measurements, and other isoprene emitters in the Arctic could be responsible for the high temperature sensitivity of isoprene.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Data are available at Wang et al. (2023).

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